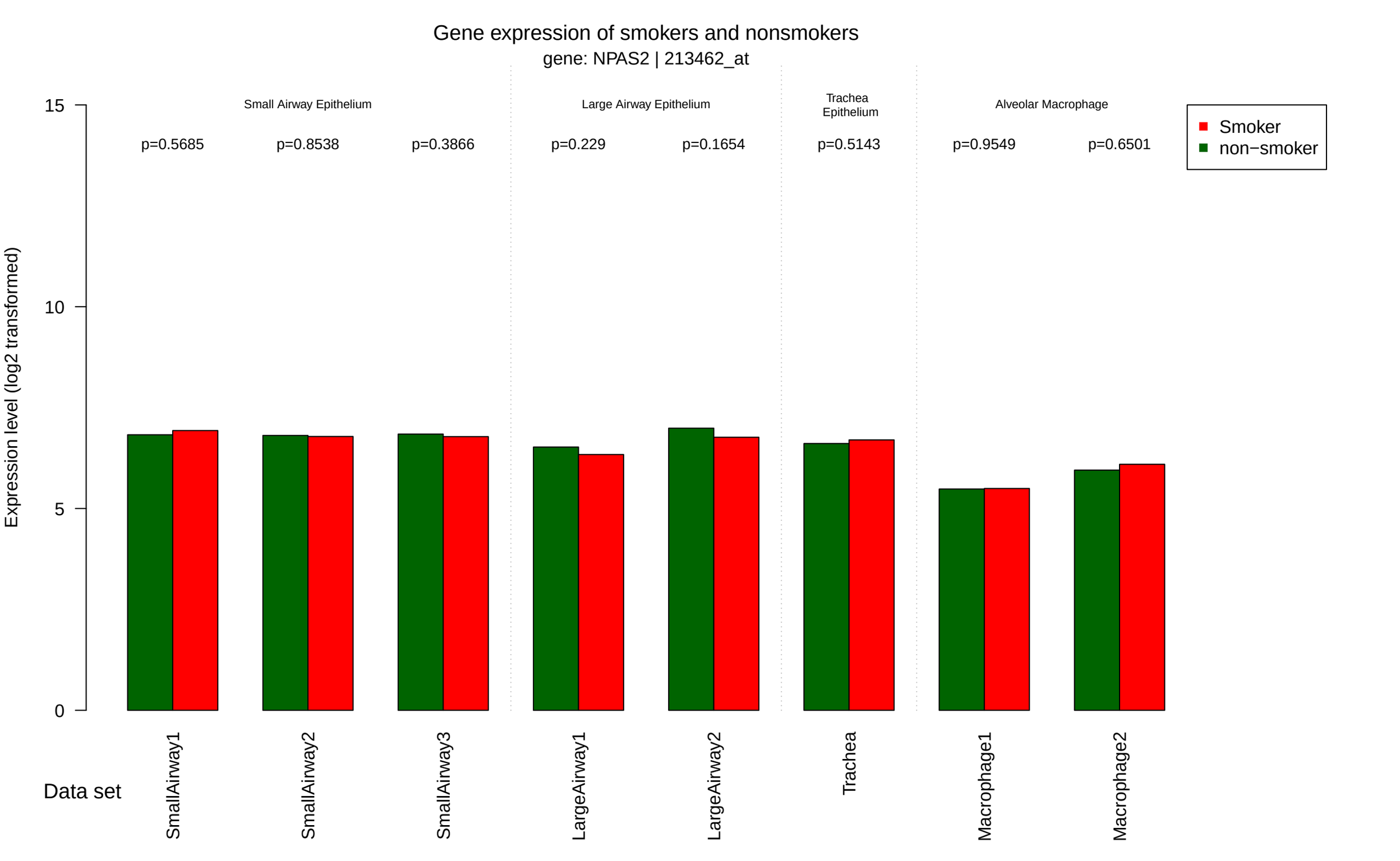
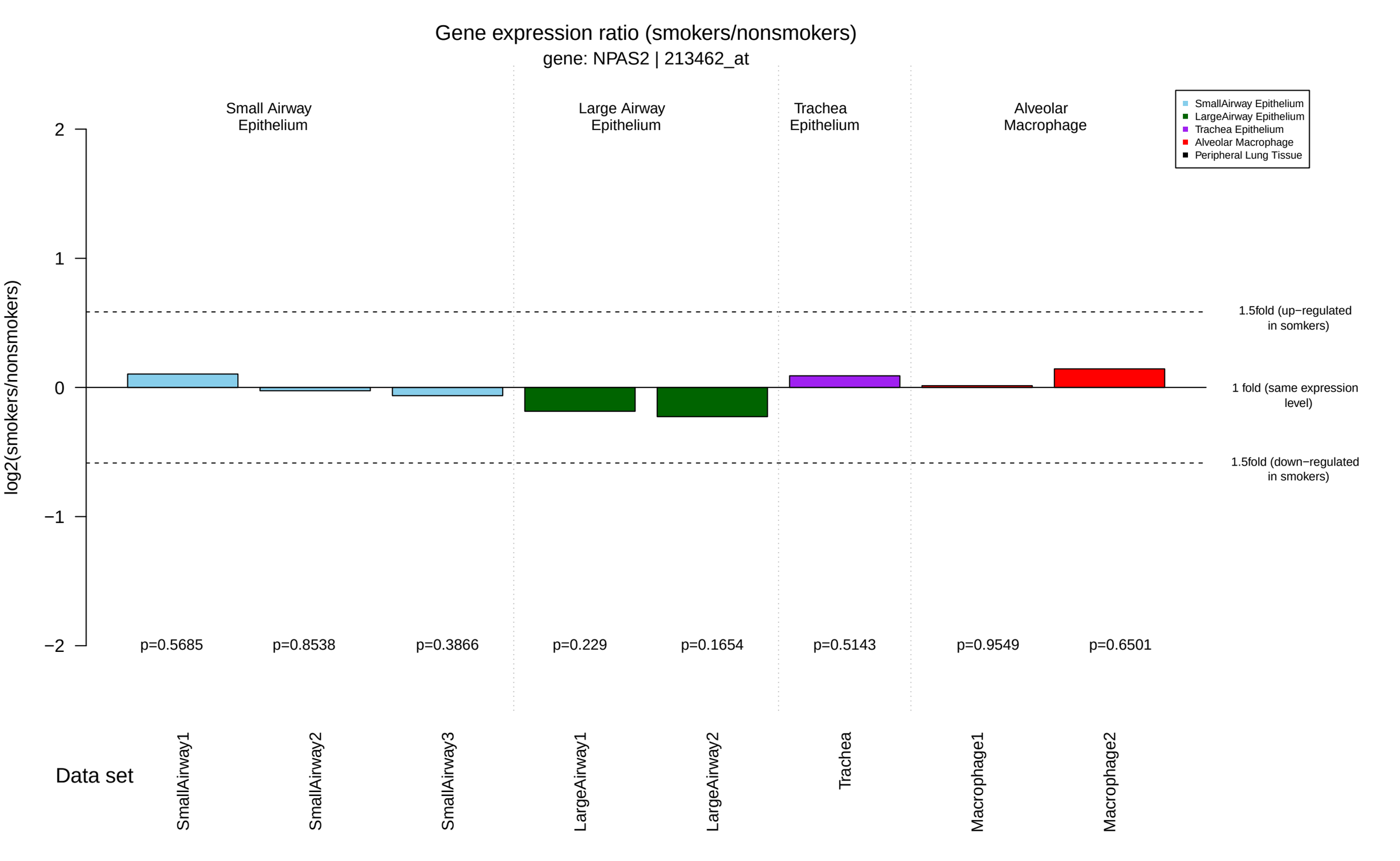
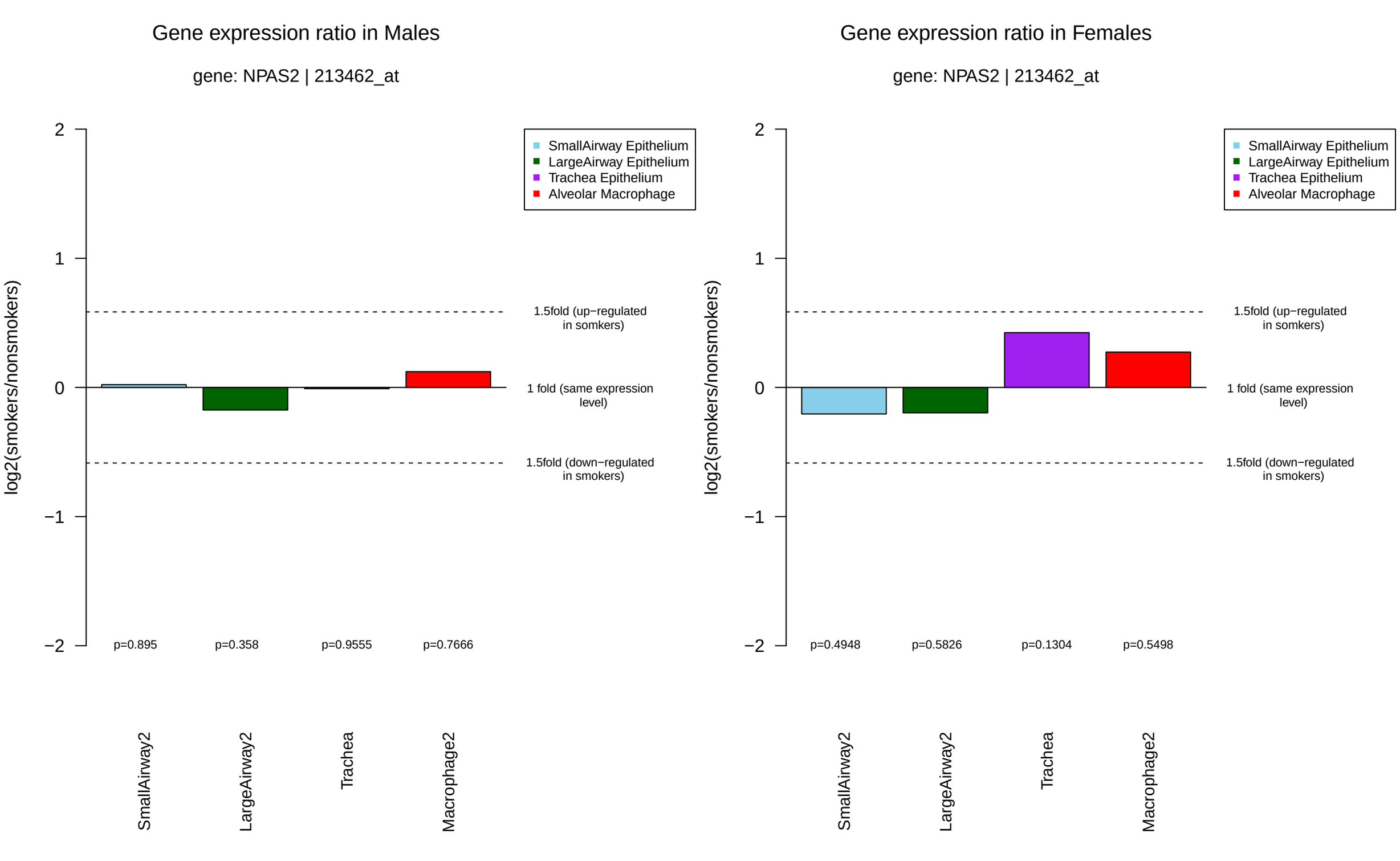
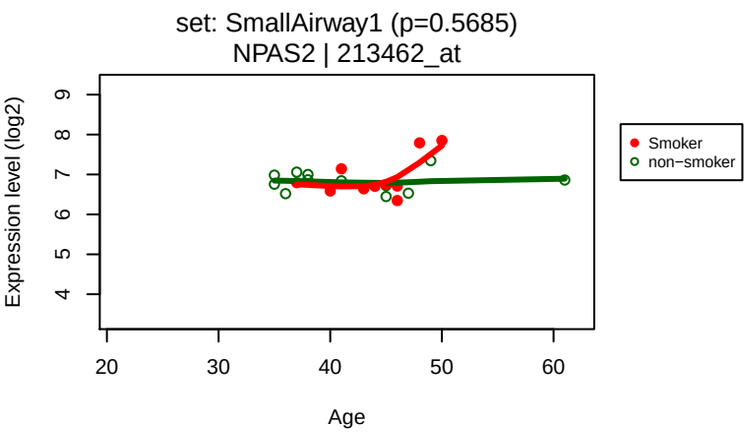
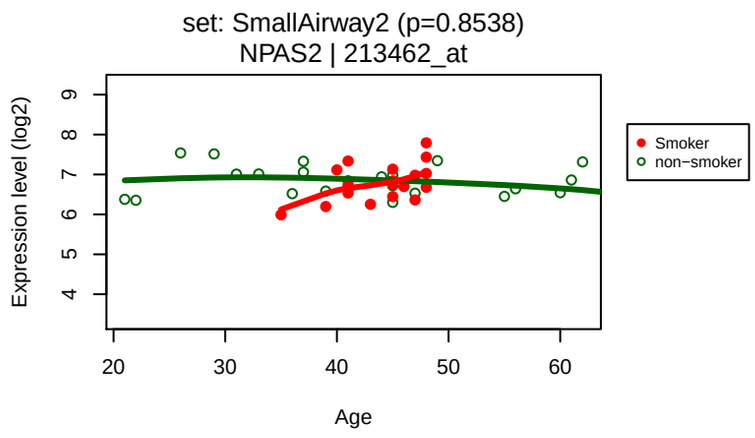
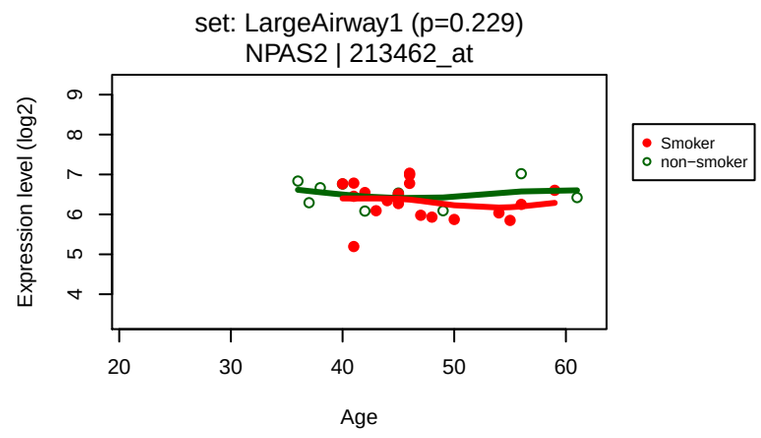

Gene expression of smokers and nonsmokers
gene: NPAS2 | 213462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5685
p=0.8538
p=0.3866
p=0.229
p=0.1654
p=0.5143
p=0.9549
p=0.6501
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

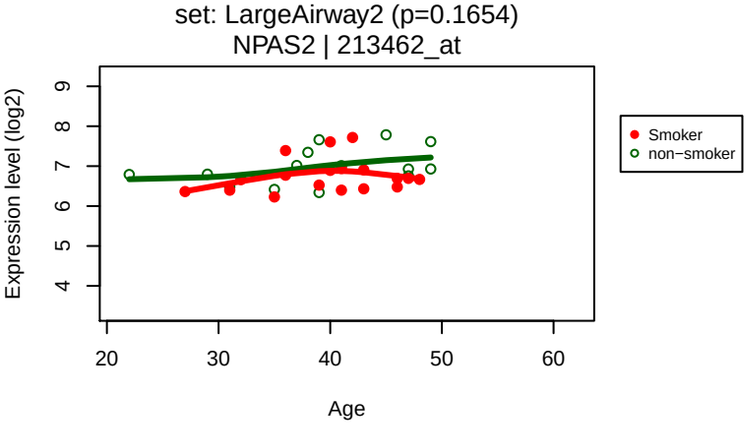

Gene expression ratio (smokers/nonsmokers)
gene: NPAS2 | 213462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5685
p=0.8538
p=0.3866
p=0.229
p=0.1654
p=0.5143
p=0.9549
p=0.6501
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

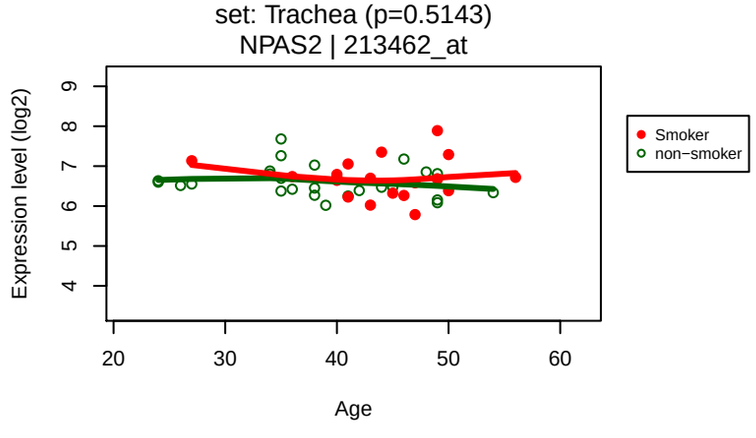

Gene expression ratio in Males
gene: NPAS2 | 213462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.895
p=0.358
p=0.9555
p=0.7666
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPAS2 | 213462_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4948
p=0.5826
p=0.1304
p=0.5498
SmallAirway2
LargeAirway2
Trachea
Macrophage2

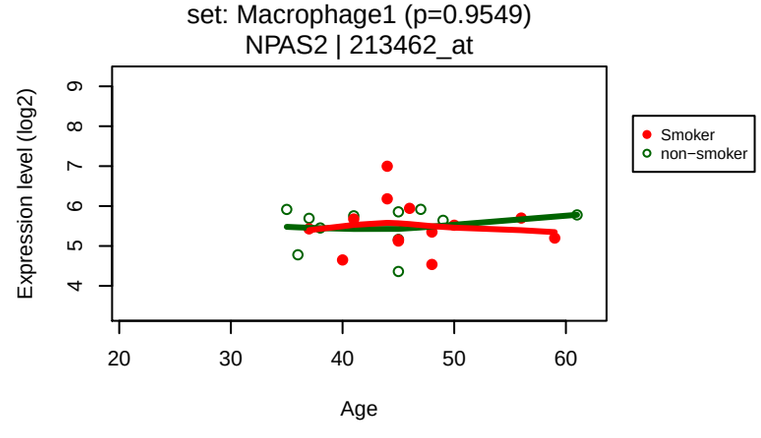

set: SmallAirway1 (p=0.5685)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker

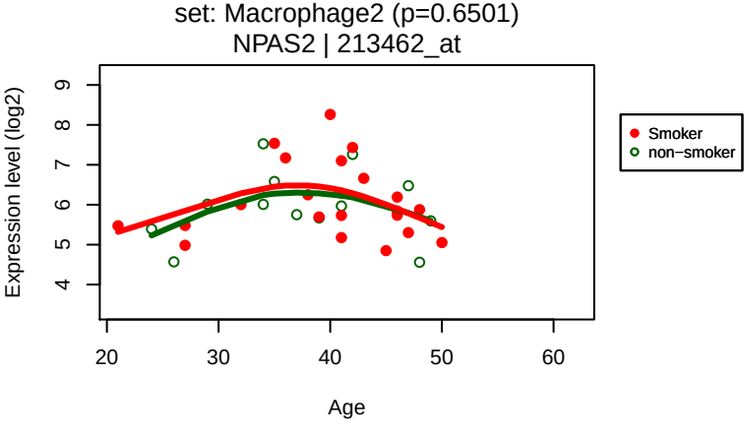

set: SmallAirway2 (p=0.8538)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker

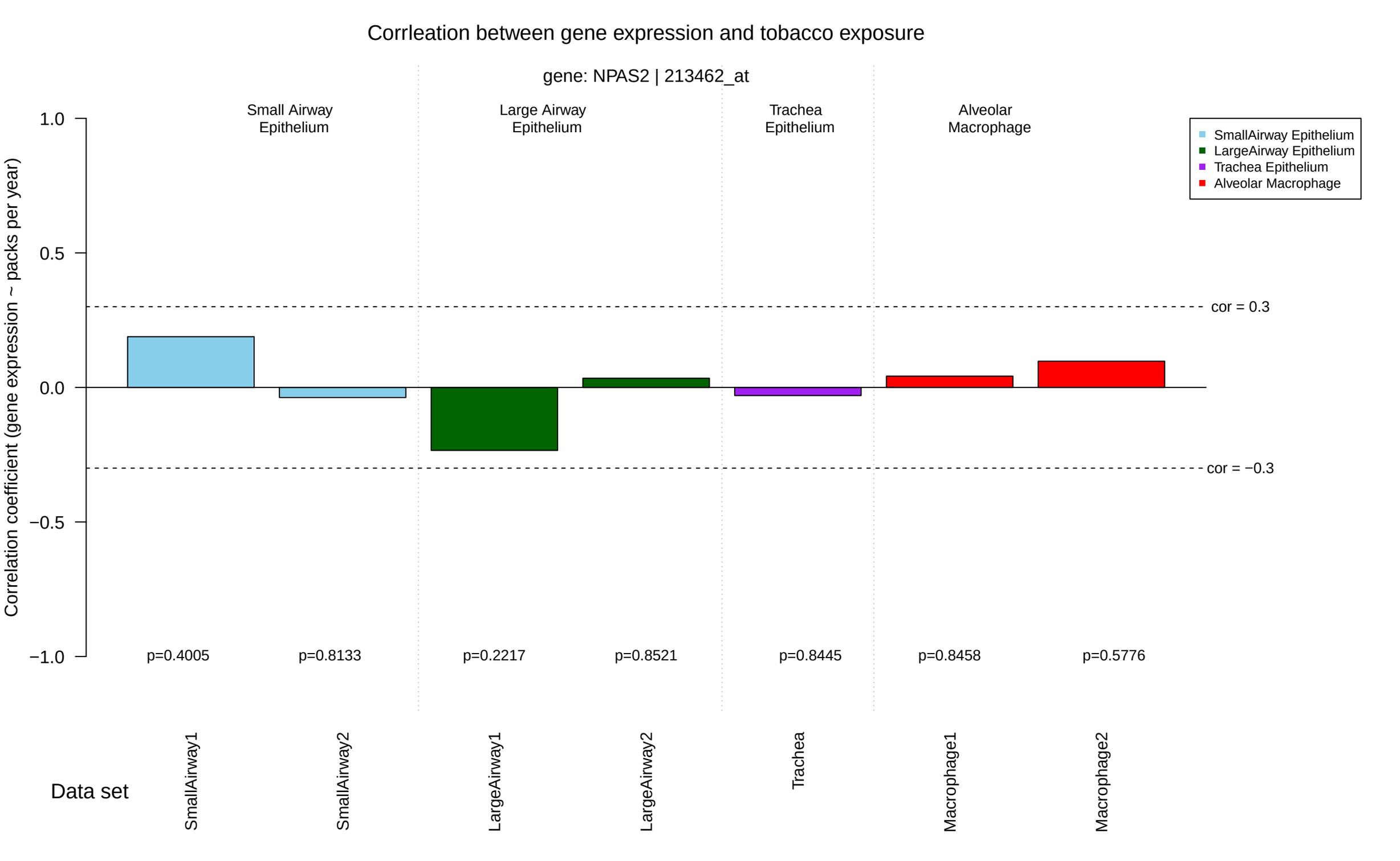

set: LargeAirway1 (p=0.229)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1654)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5143)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9549)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6501)
NPAS2 | 213462_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPAS2 | 213462_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4005
p=0.8133
p=0.2217
p=0.8521
p=0.8445
p=0.8458
p=0.5776
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2